SMART Weight Check

Weigh at source. Check every tonne.

An independent scale in the Europress compactor checks every lift and every per-tonne invoice line, and a human approves every proposal.

Since AI Hackathon 2026 · Europress challenge · all data synthetic, not Europress data

Customers pay per tonne they never see



CMP-00046 · cardboard · Lohja · Jul–Sep 2026. Top: fill % · bottom: container weight, kg (simulated). Dashed lines: trips without a lift.

July: the fill reading stuck at 88 % while the scale read ~0 kg. Four trucks came for nothing.
Nobody did anything wrong. Nobody could see it.

2.3×

€ per tonne vs the contract benchmark: €258 vs €110 (estimated)

€346

for 4 wasted trips while the fill reading was stuck (observed)

6–25 %

fill at three customer-ordered lifts in September (simulated)

Weigh, check, propose: a human decides

1 · Weigh

Four load cells under the container. Gross before the lift minus tare after = collected kg. A free zero check on every lift.

2 · Check

Every lift and every per-tonne invoice line against device weight, the operator's ticket and the contract. Combined loads and side waste are explained, not blamed.

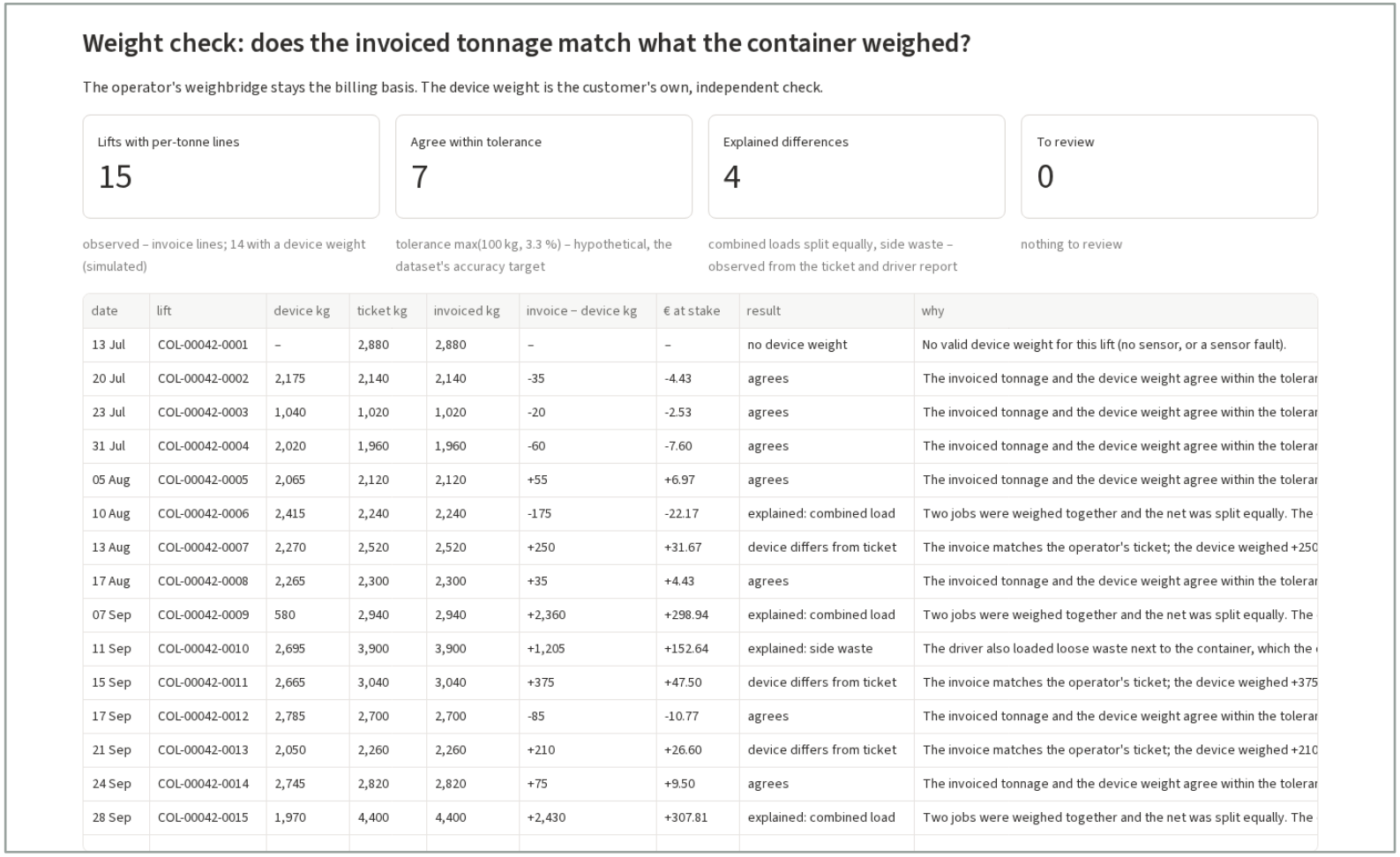
3 · Propose

Each difference becomes a proposal with a reason and a check before acting. Approve, reject, undo. Nothing executes by itself.

Every number carries its evidence:

observed read from a record	**simulated** SMART device data	**estimated** our calculation	**hypothetical** an assumption

Live demo: every lift checked



Weight check: does the invoiced tonnage match what the container weighed?

The operator's weighbridge stays the billing basis. The device weight is the customer's own, independent check.

Lifts with per-tonne lines	Agree within tolerance	Explained differences	To review
15	7	4	0
observed – invoice lines; 14 with a device weight (simulated)	tolerance max(100 kg, 3.3 %) – hypothetical, the dataset's accuracy target	combined loads split equally, side waste – observed from the ticket and driver report	nothing to review

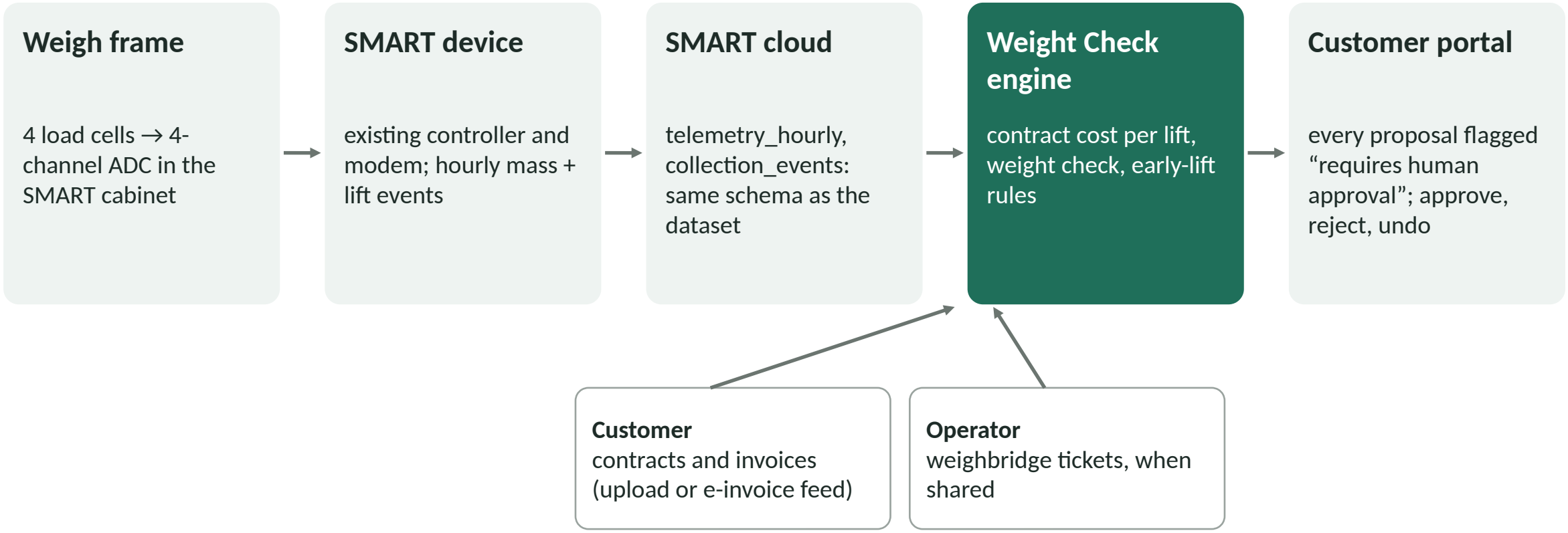
date	lift	device kg	ticket kg	invoiced kg	invoice – device kg	€ at stake	result	why
13 Jul	COL-00042-0001	–	2,880	2,880	–	–	no device weight	No valid device weight for this lift (no sensor, or a sensor fault).
20 Jul	COL-00042-0002	2,175	2,140	2,140	-35	-4.43	agrees	The invoiced tonnage and the device weight agree within the tolerar
23 Jul	COL-00042-0003	1,040	1,020	1,020	-20	-2.53	agrees	The invoiced tonnage and the device weight agree within the tolerar
31 Jul	COL-00042-0004	2,020	1,960	1,960	-60	-7.60	agrees	The invoiced tonnage and the device weight agree within the tolerar
05 Aug	COL-00042-0005	2,065	2,120	2,120	+55	+6.97	agrees	The invoiced tonnage and the device weight agree within the tolerar
10 Aug	COL-00042-0006	2,415	2,240	2,240	-175	-22.17	explained: combined load	Two jobs were weighed together and the net was split equally. The
13 Aug	COL-00042-0007	2,270	2,520	2,520	+250	+31.67	device differs from ticket	The invoice matches the operator's ticket; the device weighed +250
17 Aug	COL-00042-0008	2,265	2,300	2,300	+35	+4.43	agrees	The invoiced tonnage and the device weight agree within the tolerar
07 Sep	COL-00042-0009	580	2,940	2,940	+2,360	+298.94	explained: combined load	Two jobs were weighed together and the net was split equally. The
11 Sep	COL-00042-0010	2,695	3,900	3,900	+1,205	+152.64	explained: side waste	The driver also loaded loose waste next to the container, which the
15 Sep	COL-00042-0011	2,665	3,040	3,040	+375	+47.50	device differs from ticket	The invoice matches the operator's ticket; the device weighed +375
17 Sep	COL-00042-0012	2,785	2,700	2,700	-85	-10.77	agrees	The invoiced tonnage and the device weight agree within the tolerar
21 Sep	COL-00042-0013	2,050	2,260	2,260	+210	+26.60	device differs from ticket	The invoice matches the operator's ticket; the device weighed +210
24 Sep	COL-00042-0014	2,745	2,820	2,820	+75	+9.50	agrees	The invoiced tonnage and the device weight agree within the tolerar
28 Sep	COL-00042-0015	1,970	4,400	4,400	+2,430	+307.81	explained: combined load	Two jobs were weighed together and the net was split equally. The

Demo path

1. CMP-00046: the stuck fill line vs the flat weight line; 2.3× € per tonne
2. CMP-00042: invoiced 2.94 t, container held 580 kg. A shared truck, split equally: re-split by device weight
3. CMP-00047: no ticket, invoice 425 kg above the scale: ask for the ticket
4. Approve, then undo

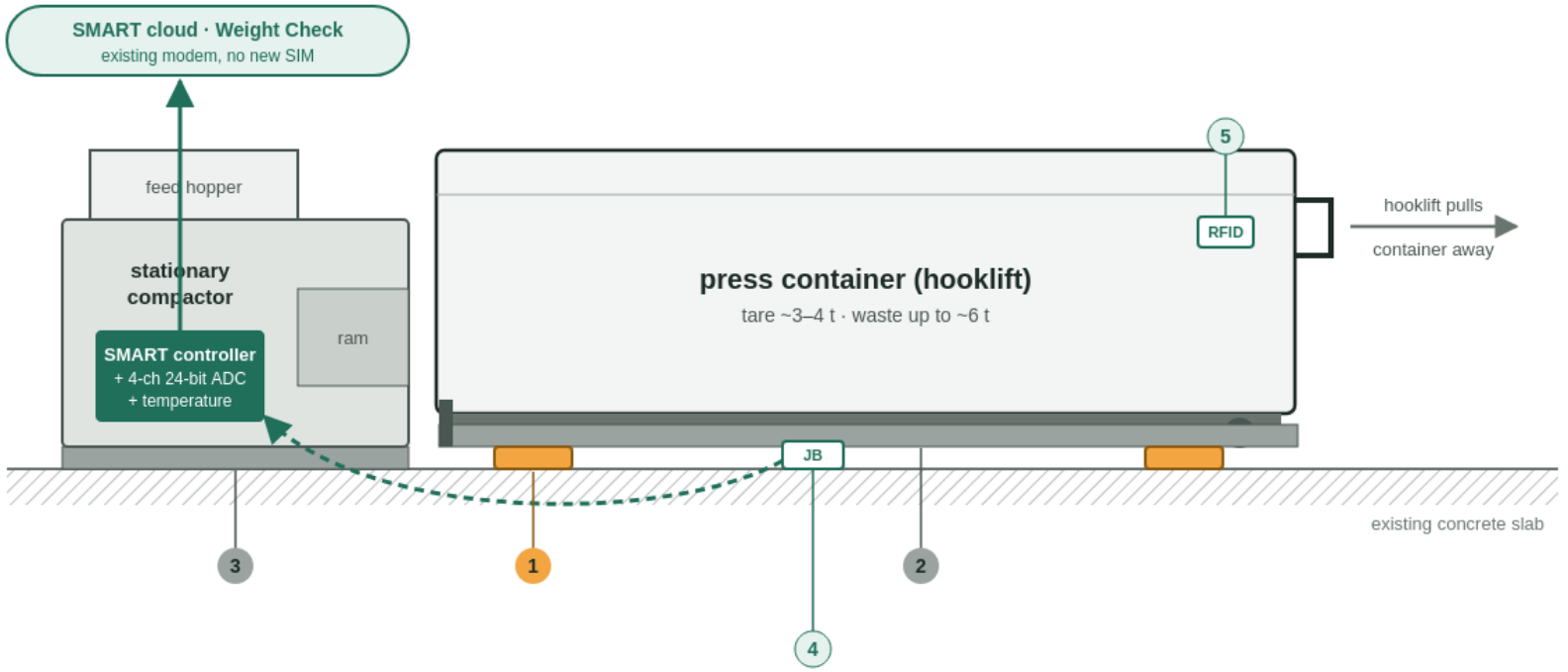
341 lifts checked: 205 agree · 38 explained · 4 billing questions · 13 devices: check the scale

From load cell to customer decision



Runs on what SMART already has: the cabinet, the modem, the telemetry pipeline. The weight is the customer's own measurement; it is shared with the operator only with consent.

A weigh frame under the container



How one lift is weighed: stable gross reading with the ram idle, minus the tare when the empty container comes back. While the container is away the frame must read zero: a free zero check on every lift. Four separate cells reveal ice or debris under one corner.

σ 1.5 %

design target; ~1.0 % accuracy budget (estimate)

€1,200

factory option · €3,400 installed retrofit (hypothetical BOM)

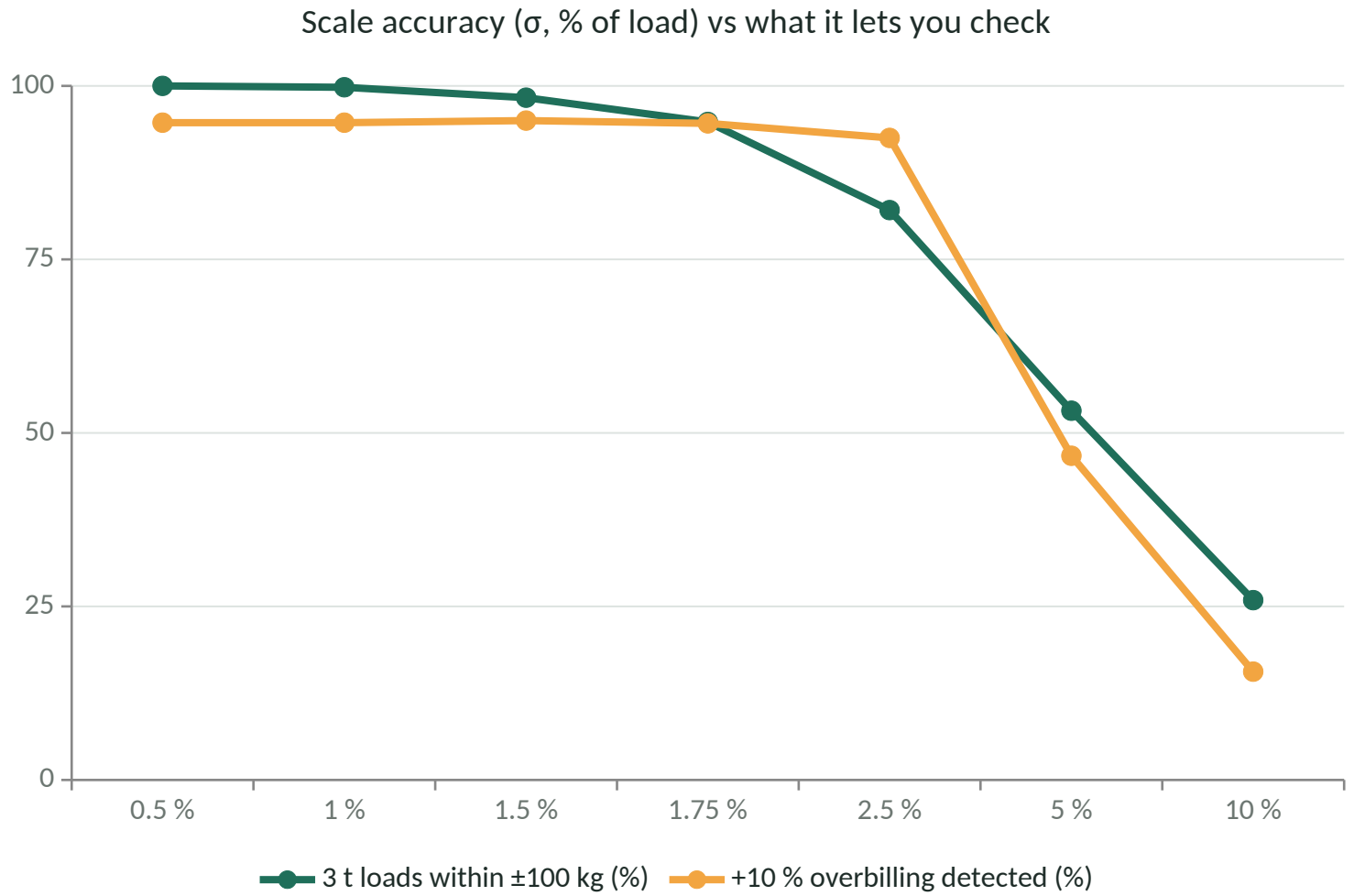
Self-checking

auto-zero every time the container leaves; span checked against the operator's tickets

Validation

2-week yard test with water tanks → 20-site pilot, 3 months → winter

Accuracy decides what weight is worth



Competition data, 10,342 shared single-load tickets; examples/weighing_accuracy_value.py (estimated)

≤ 1.75 %

keeps ≥ 95 % of 3 t loads within ±100 kg: the challenge target

> 2.5 %

and checking the operator's tonnage stops working

7 %

best estimate without a scale (ML or rotating kit): a free value scan, not an invoice check

€15 per device-month, priced from value

€36

median realisable value per device-month (estimated): the customer keeps 2.4× the fee

2.0 years

payback of onboarding at €15 and 25 % adoption (hypothetical)

Yearly profit after onboarding, € (hypothetical, 52 synthetic customers)

price / adoption	10 %	25 %	40 %
€10	943	2,358	3,773
€15	2,148	**5,370**	8,592
€25	1,735	4,338	6,941

Price matters most. €15 maximises fee revenue; at €10 small customers cost more to support than they pay (6.5-year payback).

Sell to fleets of 6+ devices

16+ devices: ~€4.0 k profit per customer-year; 2–5 devices lose money at €50 support

Weigh where it pays

factory option pays back in 2.9 years at the median device, retrofit in 8.2

Hypothesis: per-tonne full service

3 % margin ≈ €156 k/year on this sample; needs verified weight and a new role

What we have not proven, and next steps

Not proven

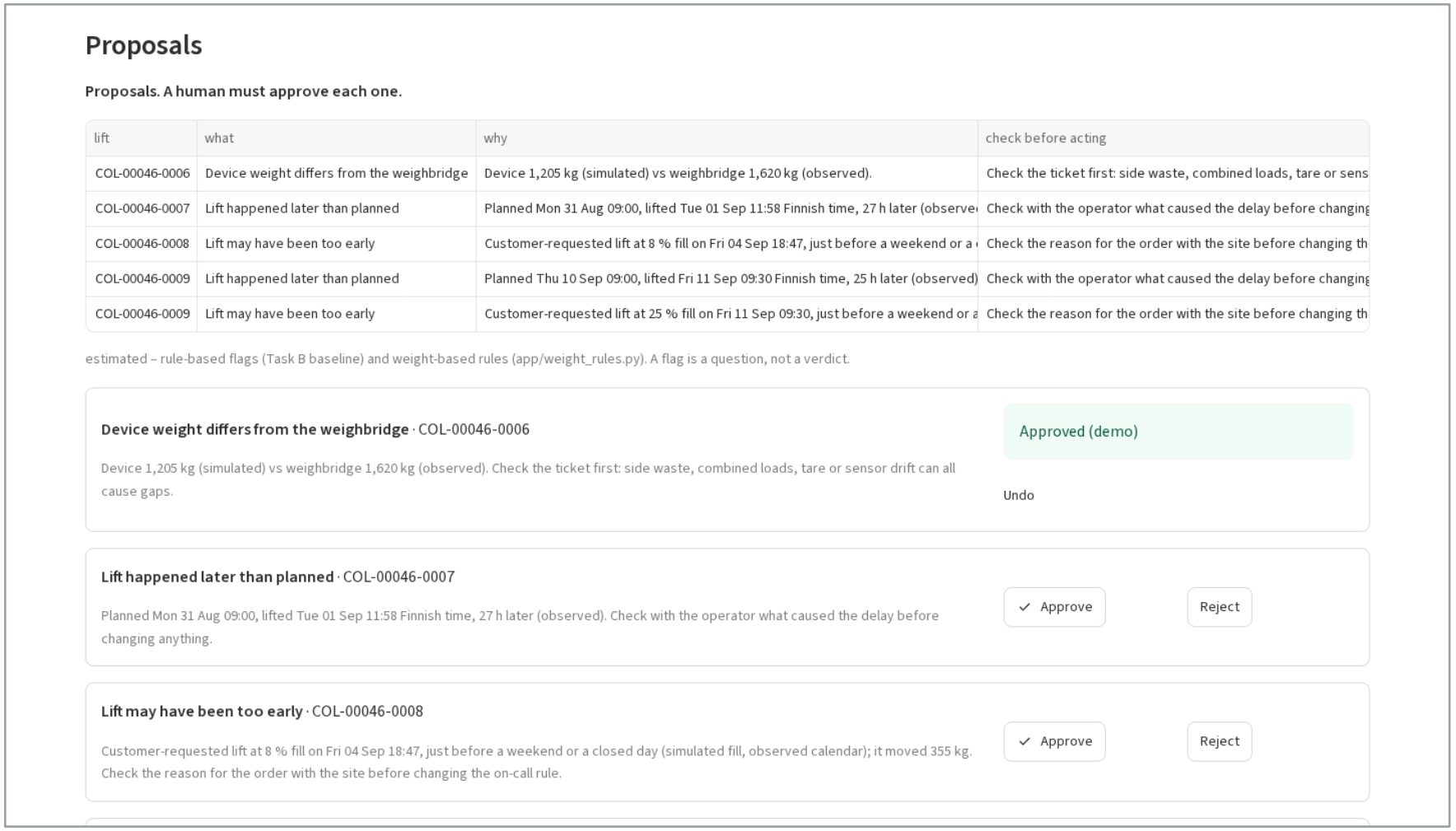
- that customers pay €15 per device-month: no interviews yet
- that the savings are real: synthetic world, hypothetical tariffs
- that the frame reaches σ 1.5 % on a real compactor in winter
- the hardware cost: indicative BOM, no supplier quotes
- that operators accept device weight as evidence

Next steps

1. 10 customer interviews: willingness to pay
2. 2-week yard test: water-tank steps, hooklift cycles
3. 20-site pilot, 3 months, vs weighbridge tickets

Weigh at source. Check every tonne.

Backup: proposals, a human decides



Proposals

Proposals. A human must approve each one.

lift	what	why	check before acting
COL-00046-0006	Device weight differs from the weighbridge	Device 1,205 kg (simulated) vs weighbridge 1,620 kg (observed).	Check the ticket first: side waste, combined loads, tare or sens
COL-00046-0007	Lift happened later than planned	Planned Mon 31 Aug 09:00, lifted Tue 01 Sep 11:58 Finnish time, 27 h later (observe	Check with the operator what caused the delay before changing
COL-00046-0008	Lift may have been too early	Customer-requested lift at 8 % fill on Fri 04 Sep 18:47, just before a weekend or a	Check the reason for the order with the site before changing th
COL-00046-0009	Lift happened later than planned	Planned Thu 10 Sep 09:00, lifted Fri 11 Sep 09:30 Finnish time, 25 h later (observed)	Check with the operator what caused the delay before changing
COL-00046-0009	Lift may have been too early	Customer-requested lift at 25 % fill on Fri 11 Sep 09:30, just before a weekend or a	Check the reason for the order with the site before changing th

estimated – rule-based flags (Task B baseline) and weight-based rules (app/weight_rules.py). A flag is a question, not a verdict.

Device weight differs from the weighbridge · COL-00046-0006

Device 1,205 kg (simulated) vs weighbridge 1,620 kg (observed). Check the ticket first: side waste, combined loads, tare or sensor drift can all cause gaps.

Approved (demo)

Undo

Lift happened later than planned · COL-00046-0007

Planned Mon 31 Aug 09:00, lifted Tue 01 Sep 11:58 Finnish time, 27 h later (observed). Check with the operator what caused the delay before changing anything.

✓ Approve Reject

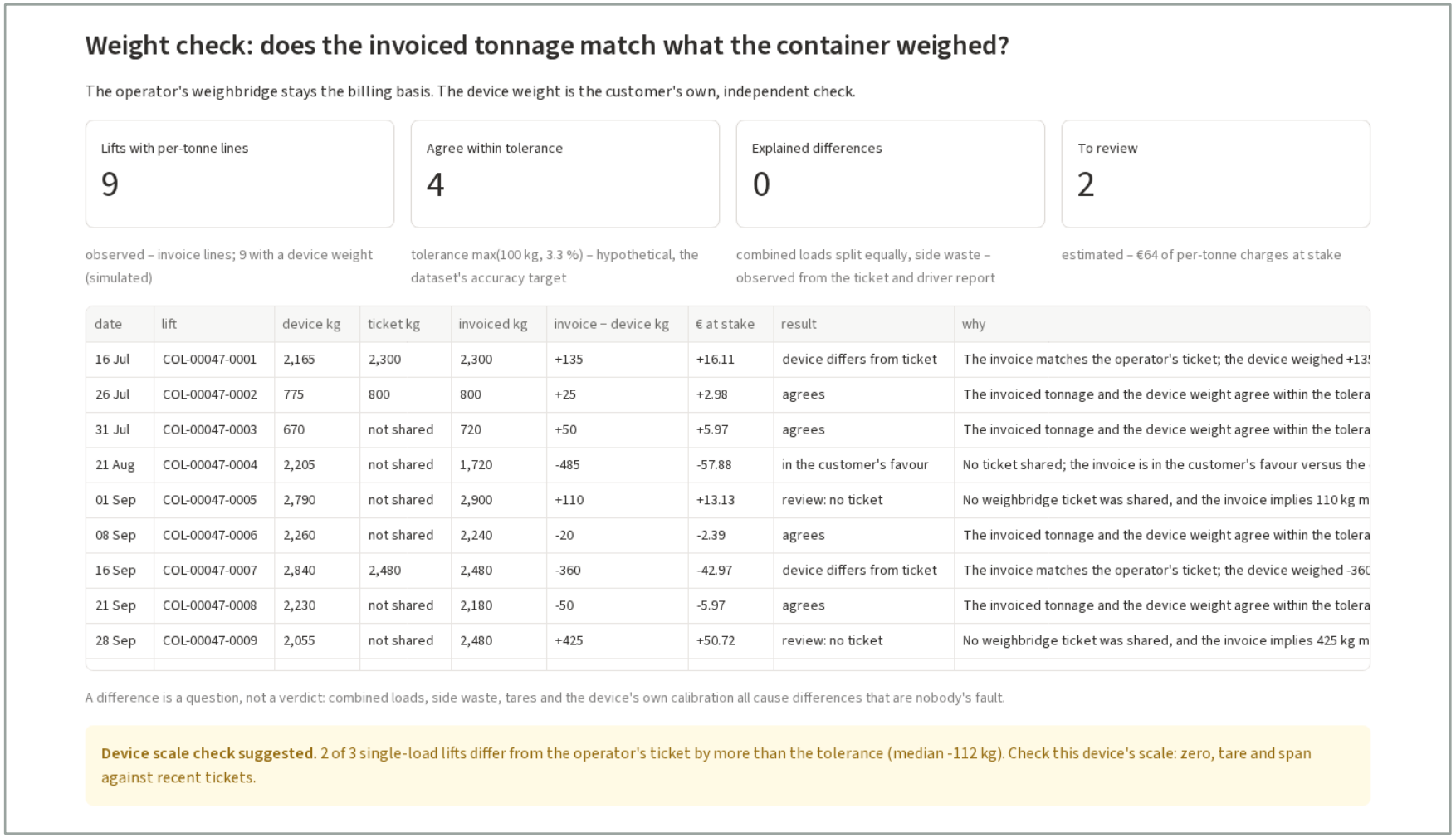
Lift may have been too early · COL-00046-0008

Customer-requested lift at 8 % fill on Fri 04 Sep 18:47, just before a weekend or a closed day (simulated fill, observed calendar); it moved 355 kg. Check the reason for the order with the site before changing the on-call rule.

✓ Approve Reject

CMP-00046: proposals with a reason and a check; one approved, with Undo. Weight rules add the early lifts on 4 and 11 September.

Backup: a lift without a ticket



Weight check: does the invoiced tonnage match what the container weighed?

The operator's weighbridge stays the billing basis. The device weight is the customer's own, independent check.

Lifts with per-tonne lines	Agree within tolerance	Explained differences	To review
9	4	0	2
observed – invoice lines; 9 with a device weight (simulated)	tolerance max(100 kg, 3.3 %) – hypothetical, the dataset's accuracy target	combined loads split equally, side waste – observed from the ticket and driver report	estimated – €64 of per-tonne charges at stake

date	lift	device kg	ticket kg	invoiced kg	invoice – device kg	€ at stake	result	why
16 Jul	COL-00047-0001	2,165	2,300	2,300	+135	+16.11	device differs from ticket	The invoice matches the operator's ticket; the device weighed +13
26 Jul	COL-00047-0002	775	800	800	+25	+2.98	agrees	The invoiced tonnage and the device weight agree within the tolera
31 Jul	COL-00047-0003	670	not shared	720	+50	+5.97	agrees	The invoiced tonnage and the device weight agree within the tolera
21 Aug	COL-00047-0004	2,205	not shared	1,720	-485	-57.88	in the customer's favour	No ticket shared; the invoice is in the customer's favour versus the
01 Sep	COL-00047-0005	2,790	not shared	2,900	+110	+13.13	review: no ticket	No weighbridge ticket was shared, and the invoice implies 110 kg m
08 Sep	COL-00047-0006	2,260	not shared	2,240	-20	-2.39	agrees	The invoiced tonnage and the device weight agree within the tolera
16 Sep	COL-00047-0007	2,840	2,480	2,480	-360	-42.97	device differs from ticket	The invoice matches the operator's ticket; the device weighed -360
21 Sep	COL-00047-0008	2,230	not shared	2,180	-50	-5.97	agrees	The invoiced tonnage and the device weight agree within the tolera
28 Sep	COL-00047-0009	2,055	not shared	2,480	+425	+50.72	review: no ticket	No weighbridge ticket was shared, and the invoice implies 425 kg m

A difference is a question, not a verdict: combined loads, side waste, tares and the device's own calibration all cause differences that are nobody's fault.

Device scale check suggested. 2 of 3 single-load lifts differ from the operator's ticket by more than the tolerance (median -112 kg). Check this device's scale: zero, tare and span against recent tickets.

CMP-00047: two lifts invoiced without a shared ticket differ from the device weight, so the proposal is to ask for the ticket. The scale-check note points at the device, not the operator.

Run it, sources and licences

Run the demo

```
python -m venv .venv && source .venv/bin/activate
pip install -e ".[dev,app]"
python baselines/b1_economic_calculator.py data/quickstart/public my_task_a.csv
python baselines/b2_rules_anomaly.py data/quickstart/public my_task_b.csv
streamlit run app/app.py
```

Where everything is

- submission/1_WEIGHING_CONCEPT.md: sensor, BOM, calibration, validation
- submission/2_SOFTWARE_DEMO.md: walkthrough, SMART integration
- submission/3_BUSINESS_CASE.md: value, pricing, sensitivity, limits
- submission/THIRD_PARTY.md: every library, dataset and service, with licence
- notes/FINDINGS.md: every number with its source table or script

Data: Europress × Since AI synthetic dataset v1.0.1 (use per the hackathon rules). Code: Python with numpy, pandas, DuckDB, Streamlit, Altair, scikit-learn, matplotlib (BSD, MIT, Apache-2.0 and PSF-style licences, all allow commercial use). AI assistance: Claude (Anthropic) via Claude Code.